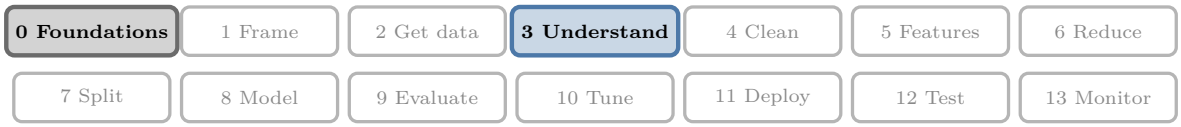
Statistics Roadmap for Machine Learning

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Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data). See the Course map.



- **Builds on:** Descriptive statistics; Inferential statistics.

1. Overview

Key point: **Statistics** (G-1884) for ML splits into four modules: **descriptive statistics** (G-596), **probability distributions** (G-1571), **inferential statistics** (G-944), and a few miscellaneous tools.

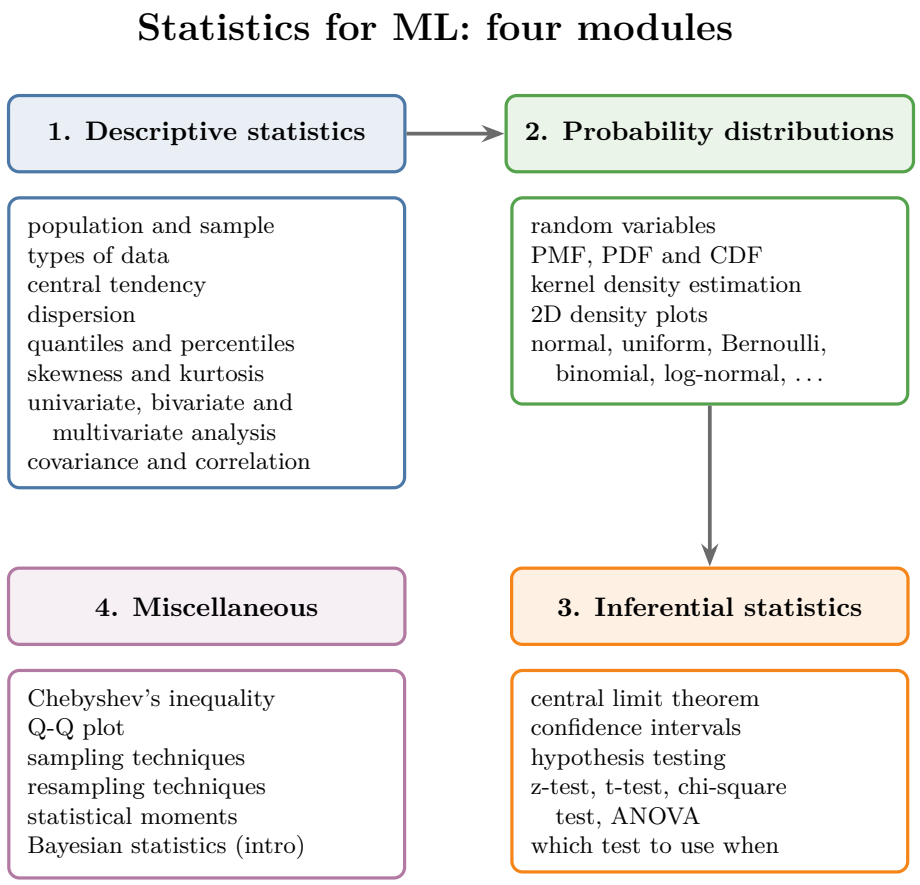


Figure 1: The four modules of statistics for ML

Figure 1 shows the whole map. Each coloured box is one module; the white box under it lists its topics. An arrow means "is needed by": descriptive statistics comes first, and probability distributions are needed by inferential statistics. **Hypothesis testing** (G-913), in the inferential module, depends on the probability distributions module. This Note says what each module covers, why ML needs it, and where each topic is taught.

2. Why ML needs statistics

> **Key point: Statistics** (G-1884) is how we summarise data and draw conclusions from it, and many ML algorithms are built on it.

Statistics turns raw data into summaries and decisions. In data work it shows up in three places (Figure 2):

- **Understanding data:** every exploratory analysis is built from statistical summaries and graphs.
- **Algorithms:** linear regression, logistic regression and Naive Bayes all come out of statistics (ISLR ch. 3, §4.3, §4.4.4).
- **Decisions:** checking whether a new medicine, or a new website design, really works better than the old one needs a statistical test (Bruce et al. 2020, ch. 3).

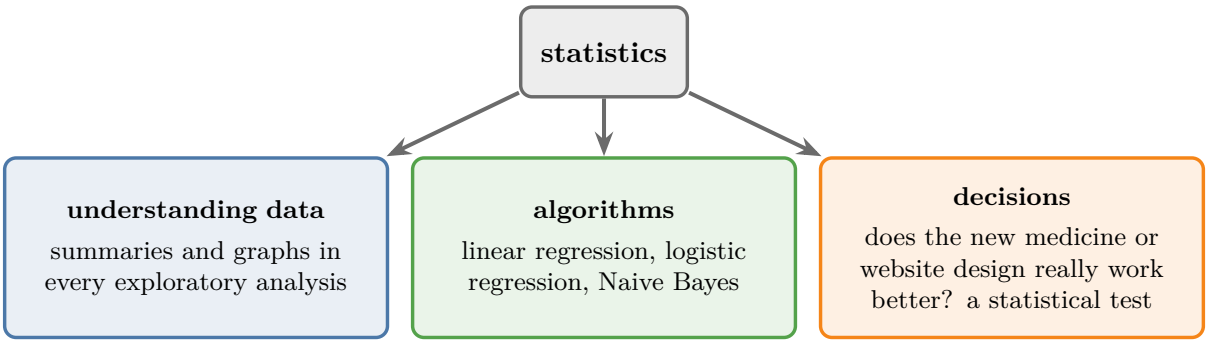


Figure 2: The three places statistics shows up in data work, with an example of each

3. The four modules

> **Key point:** Descriptive statistics summarises the data we have; **probability distributions** (G-1571) describe the shapes data can take; inferential statistics draws conclusions about data we do not have.

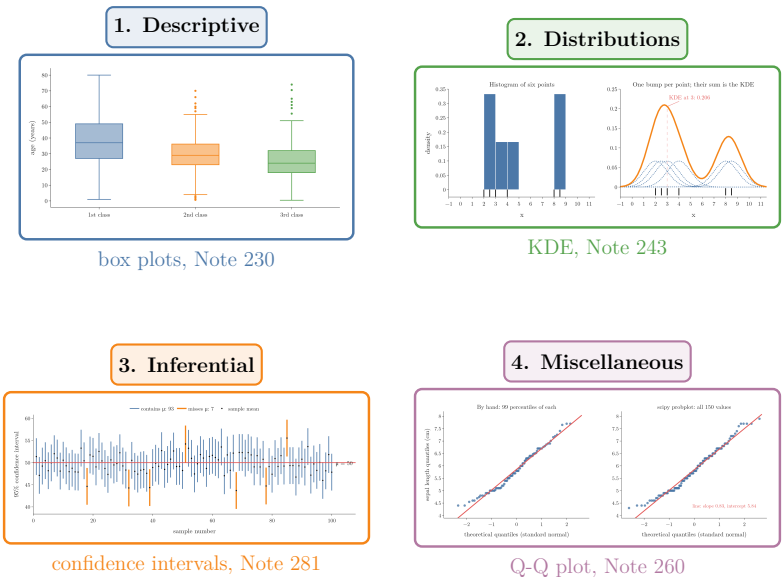


Figure 3: One key figure from each module: box plots of Titanic ages (MA-008), a kernel density estimate built from bumps (MA-023), 100 confidence intervals (MA-036) and a Q-Q plot of iris sepal lengths (MA-028)

Figure 3 previews what each module produces: summaries of the data in hand, a smooth shape for a distribution, a statement about a **population** (G-1525) with its uncertainty, and a diagnostic plot.

3.1 Descriptive statistics

> **Key point:** Summarise the data we have, with numbers and graphs.

This module describes data that is already in our hands. Its topics, and where each is taught:

Topic	Where it is taught
Population and sample, types of data	population and sample, types of data
Central tendency (mean, median, mode, . . .)	what central tendency means
Dispersion (range, variance, standard deviation, . . .)	**range** (G-1626), variance
Quantiles, percentiles, box plots	percentiles, building a box plot
Skewness	skewness as distance from the normal shape
Kurtosis	kurtosis as tailedness
Univariate analysis	one feature at a time
Bivariate and multivariate analysis	two features: scatter plot, graphs for two features
Covariance and correlation	**covariance** (G-496), **correlation** (G-490)

3.2 Probability distributions

> **Key point:** A **probability distribution** (G-1571) describes how likely each value of a feature is; the hypothesis tests of the next module depend on it.

This module studies the shapes data can take. Here a **feature** (G-772) is an input variable, one column of the data table, and an **observation** (G-1374) is one record, one row of that table:

- **Random variables** (G-1620; a number decided by chance, such as the roll of a die), and the functions that describe them: the PMF (probability mass function, the probability of each count), the PDF (probability density function, the curve for measurements) and the CDF (cumulative distribution function, the running total of probability).
- **Kernel density estimation**, the smooth curve over a histogram (a bar chart of how many values fall in each range), met in the density plot.
- **2D density plots**, the same idea for two features at once.
- **Named distributions**: normal, uniform, Bernoulli, binomial, log-normal and others. The normal distribution's 68-95-99.7 rule appears in the 68-95-99.7 rule.

The basic rules of probability are in the five basic terms and the rules every probability follows; conditional probability (the probability of one event once another is known) is in the definition.

3.3 Inferential statistics

> **Key point:** Use a sample to say something about the whole population, and say how sure we are.

This module draws conclusions about a **population** (G-1525) from a **sample** (G-1731):

- **Central limit theorem** (G-364): why averages of samples behave predictably.
- **Confidence intervals:** a range built by a method that captures the true population value in a stated share of repeated samples.
- **Hypothesis testing** (G-913): checking a claim about a population with a sample. The common tests are the z-test, t-test, chi-square test and ANOVA (Bruce et al. 2020, ch. 3), and the module ends with when to use which.

3.4 Miscellaneous topics

> **Key point:** A few tools do not fit one module; we study them when an algorithm needs them.

Topic	Where it is taught
Chebyshev's inequality	a later maths Note
Q-Q plot	how a Q-Q plot is built, checking whether a column is normal

Topic	Where it is taught
Sampling techniques	a later maths Note; sampling bias in sampling noise and sampling bias
Resampling (bootstrap)	bootstrapping
Statistical moments	**statistical moments** (G-1882)
Bayesian statistics	**Bayes' theorem** (G-269)

4. How to study the roadmap

> **Key point:** About 60 hours, or a month at 2 to 2.5 hours a day; some topics can wait until an algorithm needs them.

The whole roadmap takes roughly 60 hours. At 2 to 2.5 hours a day, that is about one month. Two habits help:

- **Know why each topic matters.** For every topic, note where ML or data analysis uses it.
- **Defer some topics.** A few topics, such as Chebyshev's inequality or Bayesian statistics, can wait until an algorithm needs them. The rest are worth covering now, in the order of the map.

> **Extra:** Good resources for this roadmap:
> - **Books:** *An Introduction to Statistical Learning* (James, Witten, Hastie, Tibshirani; **statistics** (G-1884) through ML), *Practical Statistics for Data Scientists* (Bruce, Bruce, Gedeck; practical, with Python and R), *Think Stats* (Downey; short, Python-based).
> - **Online lessons:** Starmer, J., StatQuest, *Statistics Fundamentals*, statquest.org.
> - **Revision:** a one-page statistics cheat sheet, once the topics are learned.

5. Summary

Module	Question it answers	Main topics	Why ML needs it
Descriptive statistics	What does the data we have look like?	central tendency, dispersion, percentiles, graphs, correlation	every exploratory analysis is built from these summaries and graphs
Probability distributions	What shapes can data take?	PMF, PDF, CDF, normal and other distributions	the hypothesis tests of the inferential module depend on them
Inferential statistics	What can a sample tell us about the population?	central limit theorem, confidence intervals, hypothesis tests	deciding whether a new medicine or website design really works better needs a test
Miscellaneous	Extra tools	Q-Q plot, sampling, resampling, moments, Bayes	some are studied when an algorithm needs them

- Descriptive statistics comes first; inferential statistics depends on **probability distributions** (G-1571), so study the modules in the order of the map.
- About 60 hours covers the whole roadmap, so it fits in about a month at 2 to 2.5 hours a day.
- Together the four modules are what ML needs from **statistics** (G-1884): summarise the data, describe its shape, and draw conclusions from it.

6. Sources

Built from

- CampusX, "Statistics Roadmap for Data Science and Data Analysis | Complete Guide | Full Resources | CampusX", YouTube, https://www.youtube.com/watch?v=2GV_ouHBw30

Other references

- ISLR: James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). *An Introduction to Statistical Learning*, 2nd ed. Springer. Chapter 3 (linear regression), sections 4.3 (logistic regression) and 4.4.4 (naive Bayes).
- Bruce, P., Bruce, A. and Gedeck, P. (2020). *Practical Statistics for Data Scientists*, 2nd ed. O'Reilly. Chapter 3, Statistical Experiments and Significance Testing.

7. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Statistics (G-1884)	The branch of mathematics for collecting, analysing, interpreting and presenting data.
Descriptive statistics (G-596)	Numbers that summarise data, such as count, mean, spread and percentiles.
Probability distribution (G-1571)	A list of every possible outcome of a random variable with its probability.
Inferential statistics (G-944)	Statistics that draws conclusions about a population from a sample.
Hypothesis testing (G-913)	Checking a claim about a population parameter with a sample.
Population (G-1525)	The entire group of individuals or objects we want to study.
Range (G-1626)	The simplest measure of spread: the largest value minus the smallest.
Covariance (G-496)	A number that says how two numerical features move together: positive if they rise together, negative if one rises as the other falls. Its size depends on the units.
Correlation (G-490)	A number from -1 to +1 that says how strongly two numerical features rise or fall together along a straight line: the covariance divided by both standard deviations, so the units drop out.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Random variable (G-1620)	The possible numerical outcomes of a random experiment; strictly, a function from outcomes to numbers.
Sample (G-1731)	The part of a population that we actually measure, such as 50,000 people asked about their salary instead of everyone in India; we study it because measuring the whole population is usually impossible, and use it to draw conclusions about the population.
Central limit theorem (G-364)	For large enough samples, the means of many samples follow a normal distribution centred on the population mean, whatever the shape of the data (if its variance is finite); this justifies confidence intervals and hypothesis tests on means.

Term	Meaning
Statistical moments (G-1882)	Averages of distances from the mean raised to a power: mean, variance, skewness, kurtosis.
Bayes' theorem (G-269)	The rule that reverses a conditional probability: from $P(B \mid A)$ it gives $P(A \mid B)$, so a belief about A can be updated after seeing B; $P(A \mid B) = P(B \mid A)P(A)/P(B)$.